Project Design Phase-II
Data Flow Diagram & User Stories

Date	29/09/2026
Team ID	SWTID-2026-2908
Project Name	FitSense – AI-Powered Fitness Recommendation System
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) represents the information flow within FitSense. It shows how user data enters the system, how authentication and business logic process the information, where workout data is stored, and how Gemini AI is used to generate personalized recommendations.

Level 0 DFD (FitSense):

Source	Flow	Data / Action	Destination
User	→	Register / Login / Workout Details / Fitness Inputs	Express.js Server
Express.js Server	→	Authentication & Routing	JWT / Controllers / Services
Controllers & Services	→	User & Workout Data	MongoDB / Mongoose
Controllers & Services	→	Age, Goal, Experience & Workout Statistics	Google Gemini AI
MongoDB / Gemini AI	→	Workout History / AI Recommendations / Fitness Insights	Express.js Server
Express.js Server	→	JSON Response	User

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
User	Registration	USN-1	As a user, I can register by entering my name, email and password.	Account is created successfully.	High	Sprint-1
User	Login	USN-2	As a user, I can log into the application using my email and password.	Valid credentials return a JWT token.	High	Sprint-1
User	Authentication	USN-3	As a user, I can securely access protected features.	Unauthorized requests are rejected.	High	Sprint-1
User	Workout Management	USN-4	As a user, I can add workout details such as name, category, duration, calories and date.	Workout is stored successfully.	High	Sprint-2
User	Workout History	USN-5	As a user, I can view my previous workouts.	My workout records are displayed.	High	Sprint-2
User	Workout Search	USN-6	As a user, I can search my workout history.	Matching workout records are returned.	Medium	Sprint-3
User	Workout Management	USN-7	As a user, I can update or delete my workout records.	Selected record is updated or deleted.	High	Sprint-2
User	AI Recommendations	USN-8	As a user, I can receive a personalized workout plan based on my fitness information.	Gemini generates a personalized plan.	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
User	AI Insights	USN-9	As a user, I can receive performance analysis and improvement suggestions.	AI insights are returned.	High	Sprint-3
User	AI Safety	USN-10	As a user, I can receive training tips and safety recommendations.	Relevant tips and safety guidance are returned.	Medium	Sprint-4